

Arslan Kenbayev

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EDUCATION

KAIST

Bachelor of Science in Electrical Engineering

National School of Physics and Math

High School, Math and Physics, GPA: 3.95/4.00

Aug. 2022 – Feb. 2027

Daejeon, South Korea

Sep. 2018 – May 2022

Almaty, Kazakhstan

EXPERIENCE

Edge AI Engineer Intern

KAIST IC Lab

Jan 2026 – Present

Daejeon, South Korea

- Optimized on-device ML inference latency and memory on Wear OS by deploying compact TFLite/LiteRT models under strict resource constraints.
- Achieved zero-loss signal synchronization by engineering a 50 Hz IMU data pipeline with real-time buffering and normalization techniques.
- Accelerated hardware-software integration by debugging tensor shapes, model loading, and multi-modal execution directly on Samsung Galaxy Watch 6.

Embedded Systems Engineer Intern

AXInvent

May 2025 – Mar. 2026

Daejeon, South Korea

- Accelerated smart agriculture prototyping by selecting components and designing **Altium**-based I2C/SPI/UART sensor schematics and power rails.
- Programmed the complete functional logic of each master and slave node from scratch in C to initialize, isolate, and continuously stream real-time environmental sensor data.
- Designed, integrated and validated analog/digital embedded hardware components on prototyped PCBs with power regulation and sensor signal conditioning.
- Validated field IoT sensor data integrity through oscilloscope/DMM bring-up, sensor debugging, and edge-to-cloud integration across Arduino Nano, Nvidia Jetson, Raspberry Pi 4, and server via gRPC/HTTP/2.

Semiconductor Engineer Intern

KAIST IOEL Lab

May 2025 – Aug. 2025

Daejeon, South Korea

- Optimized organic OLED performance toward lower turn-on voltage and higher luminance by fabricating CO₂-derived thin-film stacks and systematically analyzing forward-bias I-V sweep characterization data.

Automation Engineer Intern

AAEngineering

July 2024 – Sep. 2024

Almaty, Kazakhstan

- Improved gold ore segregation readiness by validating sensor integration in SolidWorks, analyzing multi-node RF/IoT topology, and prototyping automated control logic.

PROJECTS

Pipelined RISC-V CPU | *System Verilog, Verilog, ModelSim* | [\[github\]](#)

Sep. – Nov. 2025

- Designed and verified 100% testbench pass rate for a synthesizable 5-stage RISC-V pipeline CPU by implementing hazard detection/forwarding logic in SystemVerilog and debugging ModelSim waveform timing faults across IF/ID/EX/MEM/WB stages.

Ultra-Low-Power Smartwatch Firmware | *C, Arduino Nano 33 BLE, nRF5 SDK* | [\[github\]](#)

Mar. – June 2025

- Developed the complete embedded architecture and device drivers from scratch in C to achieve a DMM-verified system current draw of 85 μ A; implemented customized IMU duty cycling, step counter, an automated SSD1306 OLED display sleep routine, and coordinated active I2C sensor polling across LSM6DS and APDS-9960 hardware nodes with synchronized BLE GATT server firmware.

Digital Clock | *Verilog, EDA Playground* | [\[github\]](#)

Feb. – June 2024

- Designed and verified synchronous digital logic modules including FSMs, flip-flops, counters, sequential logic and MUXes in System Verilog/Verilog using simulation waveform analysis.

Capstone: Industrial Anomaly Detection | *Python, DCASE 2020, Git, Linux*

Mar. – June 2025

- Enabled mechanical failure detection under real-world noise by training an unsupervised autoencoder on ToyADMOS/MIMII acoustic datasets and applying reconstruction error thresholding.

TECHNICAL SKILLS

Hardware & Embedded: Wear OS, PCB Design, Soldering, Firmware Dev (nRF5 SDK, Segger Studio), Sensor Integration, I2C/SPI/UART/BLE, I-V Characterization, Oscilloscope/DMM

Languages: SystemVerilog, Verilog, C, C++, Python, MATLAB, R, Java

Platforms: Arduino Nano 33 BLE, Nvidia Jetson, Raspberry Pi, RISC-V

EDA / CAD: Altium, Fusion 360, SolidWorks, ModelSim, EDA Playground

Developer Tools: Git, VS Code, Linux